

Module Function Configuration List

Cmd No.	Function	GPIO1 Behavior	Notes
0x00	Standard Mode	Blank	
0x04	Auto inventory tags, fast switch antenna mode.	High Level On	
0x09	Auto inventory tags, fast switch antenna mode. Wiegand 26 output(in-phase).	High Level On	Suit for M-500: auto inventory tags
0x0F	Auto inventory tags, fast switch antenna mode.	Low level On	
0x11	Auto inventory tags, fast switch antenna mode. Reading Identifier of Reader at the same time.	High Level On	Intervals due to quantity of tags.
0x12	Auto inventory tags, fast switch antenna mode. Every times read tag will trigger GPIO3 output (High).	High Level On	Set trigger time by set Switch Interval between antennas, Note: unit is 20ms.
0x13	Auto inventory tags.	Low level On	Low power consumption mode. Only support 1 Ant
0x15	Auto inventory tags.Auto return GPI status.	Blank	
0x18	Auto inventory tags, fast switch antenna mode.	High Level On	Only suit for 8 port module
0x19	Auto inventory tags, Auto return the GPIO input Level	High Level On	
0x20	Auto return GPI status.	Blank	
0x24	Auto inventory tags. Only support 1 Ant	Low level On	Session S0
0x25	Auto inventory tags. Only support 1 Ant	Low level On	Session S1
0x31	Auto inventory tags, fast switch antenna mode.	High Level On	Session S1
0x32	Auto inventory tags, fast switch antenna mode.	High Level On	Session S2

- Note: 1. **GPIO1 Behavior** is just suit for Module Series;
2. Evaluation Board please see *R2000 Evaluation Board User Manual* ->2.6.1 Set GPIO Level;
3. S-8600/8800 4/8-Port Reader please see *s8600/8800 user manual*->2.6.3 Set GPIO Level.
4. Please do not send other Cmd to module frequency when module is auto working mode.

Regarding 8 Port Module pls see : **2. Setting by Cmd.**

1. Setting by software

1.1 Setting Function Configuration

Launch the supplied Software: **Function_ID.exe**, the following screen displays:

UHF RFID Function Configuration

Connection

☒ RS232 ☐ TCP/IP

RS-232

Serial Port: COM1 **Connect**

Baudrate: 115200 **Disconnect**

TCP/IP

Reader IP: 192.168.0.178 **Connect**

Port: 4001 **Disconnect**

Set function ID

Set **Get** Function ID (HEX):

Antenna Switch Sequence

Set **Get**

A	Round	B	Round	C	Round	D	Round	Switch Interval (mS)
Ant1	1	Ant2	1	Ant3	1	Ant4	1	0

Operation history:

Connecting **Reader** with **UHF RFID Function Configuration**. Selecting the corresponding **Function ID**:

UHF RFID Function Configuration

☒ RS232 ☐ TCP/IP

RS-232

Serial Port: COM3 **Connect**

Baudrate: 115200 **Disconnect**

TCP/IP

Reader IP: 192.168.0.178 **Connect**

Port: 4001 **Disconnect**

Set function ID

Set **Get** Function ID (HEX):

Antenna Switch Sequence

Set **Get**

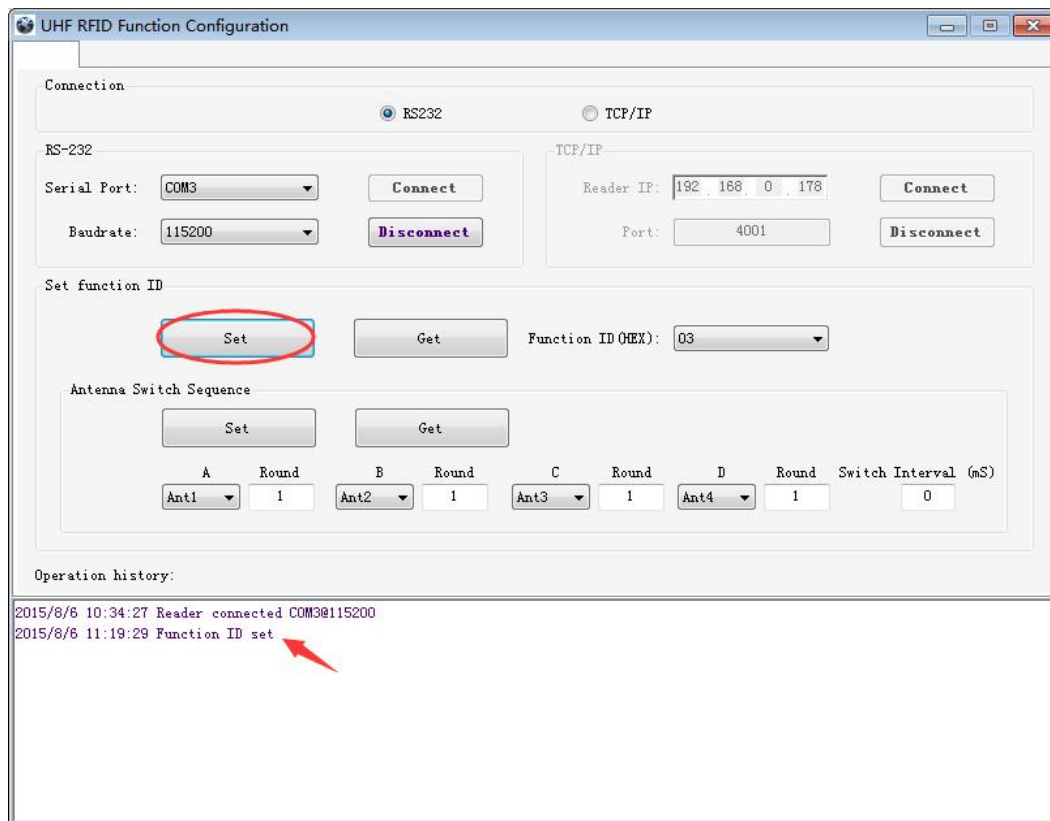
A	Round	B	Round	C	Round
Ant1	1	Ant2	1	Ant3	1

Operation history:

2015/8/6 10:34:27 Reader connected COM3@115200

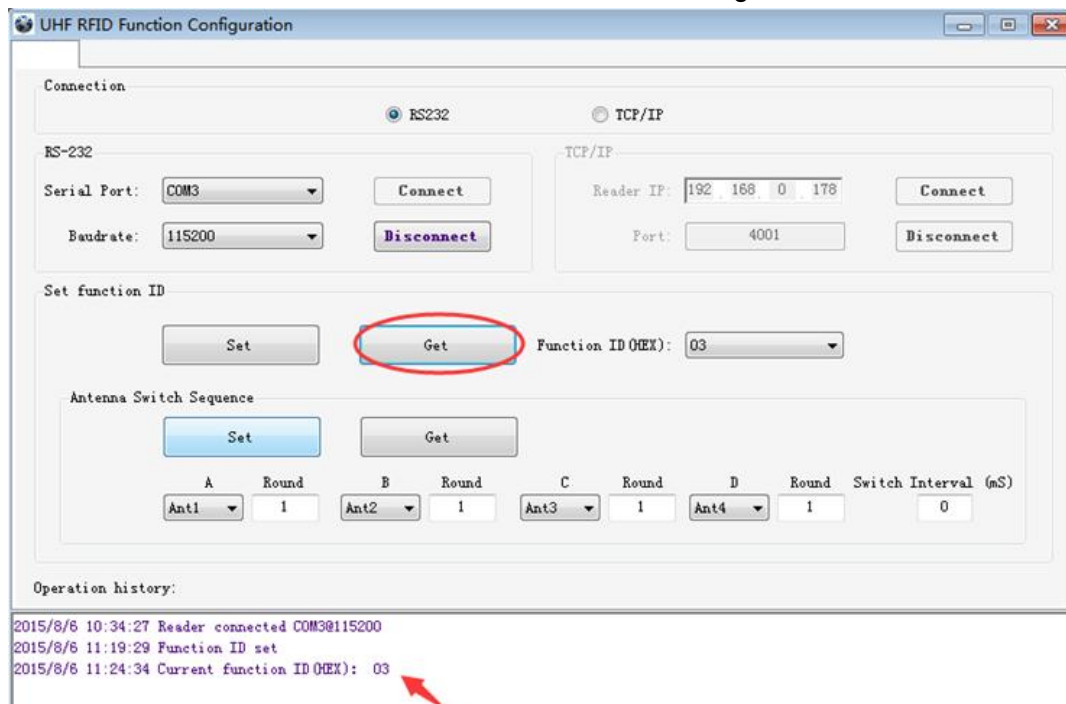
Function ID (HEX) list: 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 0A, 0B, 0C, 0D, 0E, 0F, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 1A, 1B, 1C, 1D

Click **Set**, the **Operation history** column displays:



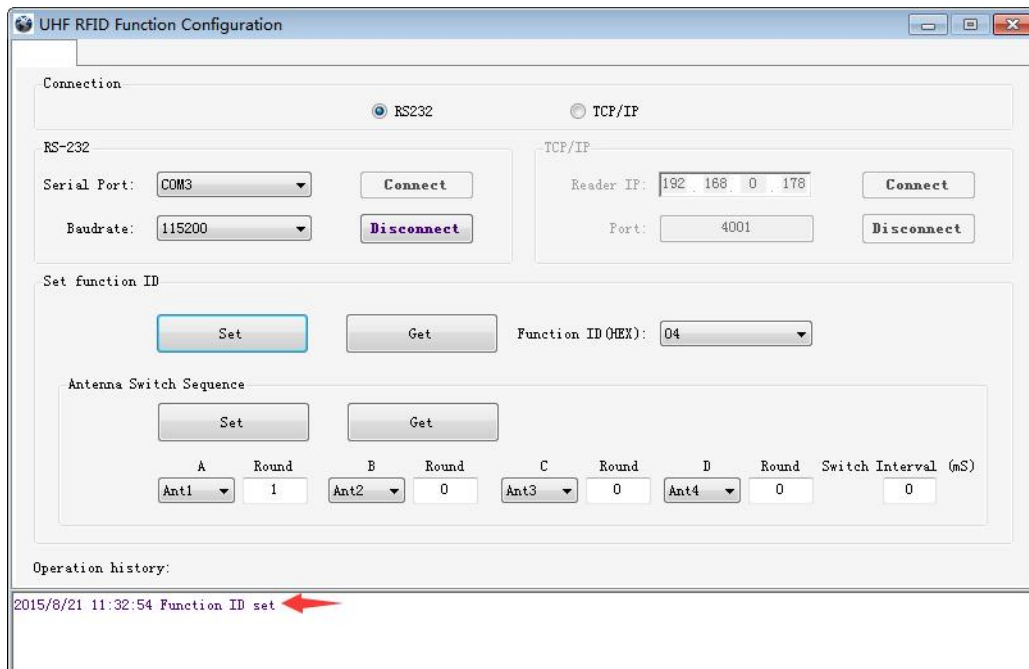
Now, **Function Configuration** has been set successfully. Reader will work as the corresponding Function Mode.

Users also can click **Get** to check the Reader working mode:

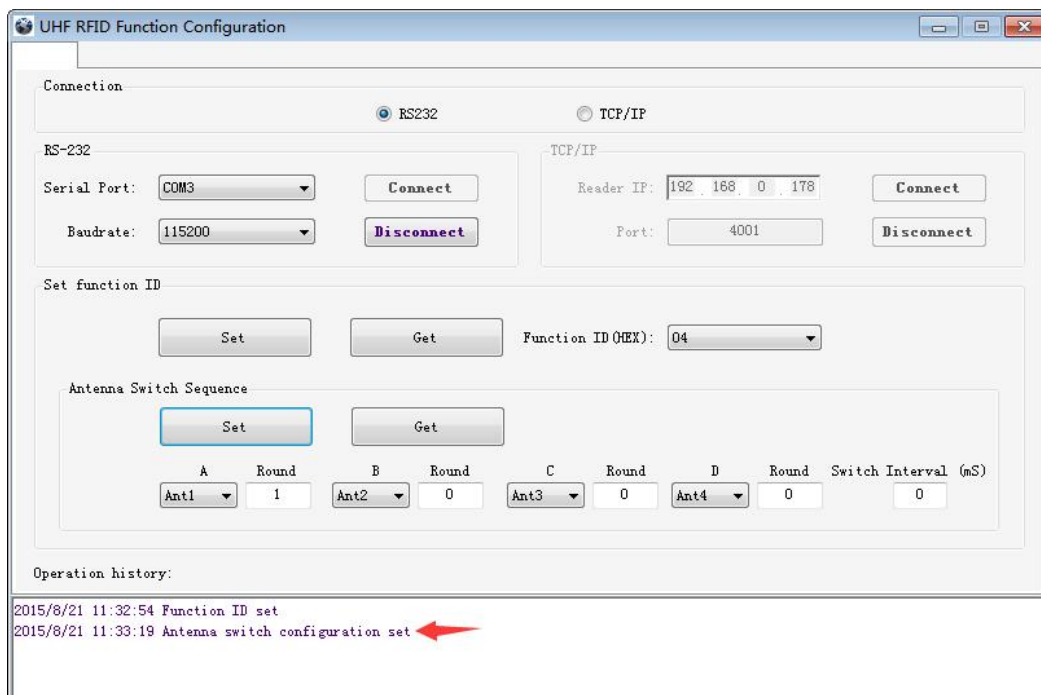


1.2 Setting Antenna Switch Sequence

For example: Setting antenna Switch Sequence under 04 mode, using Ant 1 only.
First step: Set 04 mode.



Second step: set Ant2, Ant3, Ant4 Round times to 0, click **Set**:



Then click **Get**, the follow screen as below means set **Antenna Switch Sequence** successfully:

2. Setting by Cmd.

2.1 Setting Function Configuration

Host packet:

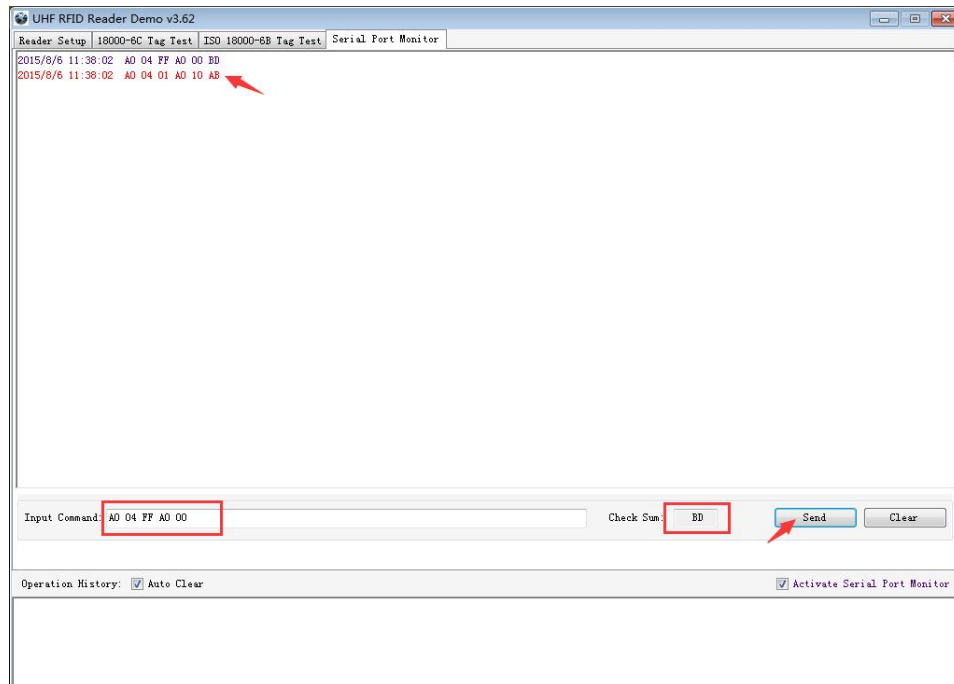
Head	Len	Address	Cmd	Check
0xA0	0x04		0xA0	

◆Succeeded:

Response packet:

Head	Len	Address	Cmd	ErrorCode	Check
0xA0	0x04		0xA0	CommandSuccess	

eg: Standard Mode: **A0 04 FF A0 00 BD**



Note: About **Check** please see *UHF RFID Reader Serial Interface Protocol V3.1..*
Then, Reader will work as the corresponding Function Mode.

2.2 Setting Antenna Switch Sequence

Cmd of Setting Antenna Switch Sequence is **cmd_fast_switch_ant_inventory**:

Host packet:

Head	Len	Address	Cmd	A	Stay	B	Stay	C	Stay	D	Stay	Interval	Repeat	Check
0xA0	0x0D		0x8A											
Parameter Description	A		First working ant (00 – 03). If set this byte above 03 means ignore it.											
	Stay		Inventory round for an antenna. Every antenna has this parameter.											
	B		Second working ant (00 – 03). If set this byte above 03 means ignore it.											
	C		Third working ant (00 – 03). If set this byte above 03 means ignore it.											
	D		Fourth working ant (00 – 03). If set this byte above 03 means ignore it.											
	Interval		Rest time between switching antennas. During the cause of rest, RF output will be cancelled, thus power consumption and heat generation are both reduced.											
	Repeat		Repeat the inventory with above ant switch sequence.											

For example: Setting antenna Switch Sequence under 04 mode, using Ant 1 only.

1. First step, set 04 mode, send cmd: **A0 04 FF A0 04 B9**
2. Second step, send cmd: **A0 0D FF 8A 00 01 01 00 02 00 03 00 00 01 C2**

Then, Ant1 of module will auto working under 04 mode.

Note: 8 Port Module pls see *M-2900 Communication Interface Specification*